

FEB22009-25: Financial Case Study
Specialisation: Econometrics & Quantitative Finance
Volatility in Real Time: Leverage, Feedback, or Both?

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1 Introduction

Equity markets show a clear pattern: large negative returns often go together with sharp rises in volatility. The pattern is easy to see, but harder to explain. Does volatility rise because prices have fallen, or do prices fall because volatility has risen? This case study asks whether the effect is mainly due to leverage, feedback, or both.

The *leverage effect* says that negative returns raise future volatility more than equally large positive returns. Here, returns move first: a bad return today raises volatility tomorrow. The *volatility-feedback effect* works in the other direction. If volatility rises, investors may demand more compensation for risk, which can push prices down at once. The key question is whether returns drive volatility, or the other way around. This question is motivated by ongoing work by Kleen, Lange, and Smetanina ([Kleen et al., 2025](#)).

A widely used market measure of volatility is the VIX, published by the Chicago Board Options Exchange (CBOE). It is based on option prices and measures expected S&P 500 volatility over roughly the next month. As a forward-looking measure, the VIX is a useful market benchmark. Figure 1 shows the VIX over time and its link with returns. The right panel shows a clear negative same-day relation: VIX increases tend to coincide with negative returns. This fits a volatility-feedback story, but is not conclusive by itself.

To evaluate volatility forecasts, you will use realised variance: the five-minute series `rv5_ss` from the [VOLARE](#) database of [Cipollini et al. \(2026\)](#). Realised variance is built from high-frequency intraday returns and is an *ex post* proxy for daily volatility. It is useful for forecast evaluation because it is observed after the trading day, although it is still a noisy proxy for latent volatility. The VIX is forward-looking and market-based. Thus, realised variance is used as the “truth” for evaluation, while the VIX is an external forecast benchmark.

The models represent different economic stories. Classic generalized autoregressive conditional heteroskedasticity (GARCH) models use past returns to update volatility. In these models, today’s return shock affects future volatility, which fits a leverage story. Real-time GARCH (RT-GARCH) models add a same-day channel: today’s return and today’s volatility can be linked within the same trading day. This helps them capture volatility feedback. In both model families, you will also study asymmetric versions, where negative and positive shocks can have different effects. This idea goes back to Glosten, Jagannathan and Runkle (GJR, [1993](#)).

You will compare four main models: GARCH, GJR-GARCH, RT-GARCH, and RT-GJR-GARCH. You will evaluate them against two benchmarks: the heterogeneous autoregressive realised variance (HAR-RV) model ([Corsi, 2009](#)) and the VIX.

These ideas lead to the following research questions:

1. Do RT-GARCH models produce more accurate one-day-ahead volatility forecasts for the S&P 500 than classic GARCH-type models?

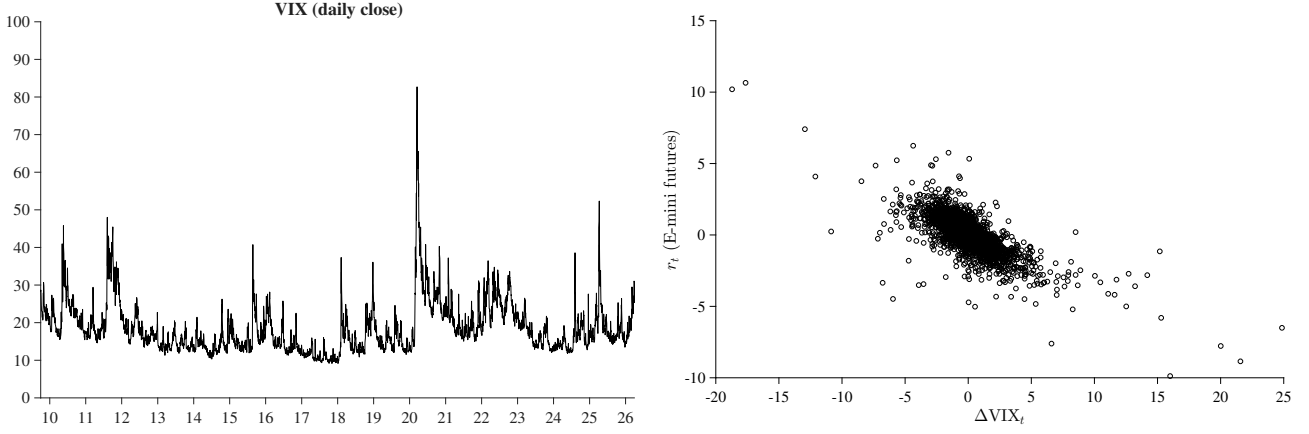


Figure 1: Left panel: the VIX over the sample period. Right panel: the negative contemporaneous relation between VIX changes (horizontal) and close-to-close returns on the E-mini S&P 500 futures contract (vertical).

2. Do asymmetric specifications improve forecasting performance within both model families?
3. What do the results suggest about the relative importance of leverage effects and volatility-feedback effects in equity-market volatility?

We focus on one-day-ahead forecasts. This keeps the theory manageable while still letting you study the timing of returns and volatility. Your main tasks are to describe the data, filter volatility, estimate the models, make forecasts, and compare them. Your report should not only present the *numbers*; it should use them to support one of three *stories*: leverage, feedback, or both.

Creativity. The case study has a clear baseline approach, but you may go beyond it. You may add another model, use different loss functions, or study extra assets. If you do so, explain why your choices are useful and how they help answer the main question, or a clear extension of it. Creative extensions can strengthen your report, but they also add risk. Please discuss major changes with us during the feedback sessions.

2 Data

All groups should use `data.xlsx` for the baseline analysis. The dataset contains daily observations for the E-mini S&P 500 futures contract, matched with realised variance and VIX data. The sample has 4,179 trading days, from 30 September 2009 to 31 March 2026.

The file contains daily opening and closing prices for the E-mini S&P 500 futures contract. The close-to-close log return is

$$r_t = 100 \log \left(\frac{\text{Close}_t}{\text{Close}_{t-1}} \right),$$

so returns are measured in percent. The first return is missing because there is no previous closing price. After dropping it, the return sample contains 4,178 observations. Later sections may use fewer observations because of lags, initialisation, or forecast evaluation windows.

The realised-variance variable is $\text{RV5_SS} \times 10^4$, taken from the [VOLARE](#) database of [Cipollini et al. \(2026\)](#). Multiplying by 10^4 puts realised variance on the same squared-percent scale as r_t^2 . We write this scaled series as rv5_ss_t . The file also contains the daily VIX close, downloaded from the official [CBOE VIX historical data page](#).

In your report, describe the data as in an empirical study, as if you gathered it yourselves. Report the sample period, number of observations, variables, units, and data sources. Explain how returns are computed and why realised variance is scaled. At a minimum, plot returns, realised variance, and the VIX, and report summary statistics such as the mean, standard deviation, minimum, maximum, skewness, and kurtosis.

Base the main analysis on `data.xlsx`. Extensions may use individual stocks, exchange rates, or other futures contracts. These data have also been provided or can be downloaded from the [VOLARE](#) database.

Demo code: The MATLAB file `Demo.m` plots the data and estimates a simple GARCH model. Use it as a starting point. The real-time models require extra coding.

3 Classic GARCH-type models

We start with classic GARCH models. They provide the lagged-volatility setup against which the real-time models will be compared. Let r_t be the close-to-close log return defined above. A standard starting point is

$$r_t = \mu + \sqrt{h_t} \varepsilon_t, \quad 1 \leq t \leq T, \quad (3.1)$$

where μ is the average return, T is the sample size, and ε_t is an i.i.d. shock with mean zero and variance one. Throughout the assignment, you may assume $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, 1)$. The process h_t is the model-implied conditional variance of returns. Since returns are in percent, h_t is in squared percentage points. We use “volatility” loosely for both standard deviation and variance.

In classic GARCH-type models, h_t is predictable at time $t - 1$: once the parameters and starting value are fixed, it can be computed from past returns. Thus, h_t is known given the information set $\mathcal{I}_{t-1}$, which contains returns up to r_{t-1} . Once h_t is known, the shock is

$$\varepsilon_t = \frac{r_t - \mu}{\sqrt{h_t}}. \quad (3.2)$$

The next two subsections introduce the symmetric GARCH filter of [Bollerslev \(1986\)](#) and its asymmetric Glosten–Jagannathan–Runkle GARCH (GJR-GARCH) extension; see also Chapter 7 of [Franses et al. \(2014\)](#).

3.1 GARCH

The standard GARCH model is

$$\begin{aligned} h_{t+1} &= \omega + \alpha(r_t - \mu)^2 + \beta h_t, \\ &= \omega + \alpha h_t \varepsilon_t^2 + \beta h_t, \end{aligned} \quad (3.3)$$

where $\omega, \alpha, \beta \geq 0$ are static parameters to be estimated. Here, ω controls the long-run variance level, α measures the short-run reaction to return shocks, and β measures persistence. Everything on the right-hand side is known at time t , so h_{t+1} can be computed step by step. Typically, $\beta \approx 0.8$, so high volatility today tends to be followed by high volatility tomorrow. The unconditional variance of r_t equals the unconditional mean of h_t :

$$E[(r_t - \mu)^2] = E[h_t \varepsilon_t^2] = E[h_t]E[\varepsilon_t^2] = E[h_t] = \bar{h},$$

where $\bar{h} := E[h_t]$. Taking expectations in (3.3) and rearranging gives

$$\bar{h} = \frac{\omega}{1 - \alpha - \beta}, \quad (3.4)$$

provided that $\alpha + \beta < 1$. A common starting value for the filter is $h_1 = \bar{h}$.

3.2 GJR-GARCH

The leverage effect says that negative shocks often raise future volatility more than positive shocks of the same size. To capture this asymmetry, we use the GJR-GARCH model of [Glosten et al. \(1993\)](#):

$$h_{t+1} = \omega + \alpha_t h_t \varepsilon_t^2 + \beta h_t, \quad (3.5)$$

where

$$\alpha_t := \alpha_1 \mathbb{1}_{r_t \leq \mu} + \alpha_2 \mathbb{1}_{r_t > \mu}. \quad (3.6)$$

The indicator $\mathbb{1}_A$ equals one if event A is true, and zero otherwise. Thus, $\alpha_t = \alpha_1$ after a negative shock and $\alpha_t = \alpha_2$ after a positive shock. The leverage effect is measured by the difference between α_1 and α_2 . Because $r_t - \mu = \sqrt{h_t} \varepsilon_t$ and $h_t > 0$, the signs of $r_t - \mu$ and ε_t are the same. If $\alpha_1 > \alpha_2$, negative returns increase next-day volatility more than positive returns. Symmetric GARCH is recovered when $\alpha_1 = \alpha_2 = \alpha$.

Because the Gaussian shock distribution is symmetric, define $\bar{\alpha} := (\alpha_1 + \alpha_2)/2$ as the average impact of squared returns. Then the unconditional mean of h_t can again be written as

$$\bar{h} = \frac{\omega}{1 - \bar{\alpha} - \beta},$$

as long as $\bar{\alpha} + \beta < 1$. As before, a common initialisation is $h_1 = \bar{h}$.

4 Real-time GARCH and volatility feedback

We now introduce real-time GARCH (RT-GARCH) models. Classic GARCH-type models build today's volatility from information available up to yesterday. RT-GARCH adds a same-day feedback channel, so today's volatility may depend on today's return shock. The cost is that h_t can only be filtered after observing r_t . The specification below is related to [Smetanina \(2017\)](#) and [Ding \(2023\)](#). It comes from ongoing joint work by Kleen, Lange, and Smetanina, which you should cite as [Kleen et al. \(2025\)](#).

4.1 Model idea and specification

The return equation is again

$$r_t = \mu + \sqrt{h_t} \varepsilon_t, \quad (4.1)$$

where ε_t is i.i.d. with mean zero and variance one, and is assumed Gaussian throughout the assignment. The RT-GJR-GARCH volatility recursion is

$$h_t = \omega + \beta h_{t-1} + \alpha_{t-1} h_{t-1} \varepsilon_{t-1}^2 + \phi_t h_{t-1} \varepsilon_t^2, \quad (4.2)$$

with

$$\alpha_t := \alpha_1 \mathbb{1}_{r_t \leq \mu} + \alpha_2 \mathbb{1}_{r_t > \mu}, \quad \phi_t := \phi_1 \mathbb{1}_{r_t \leq \mu} + \phi_2 \mathbb{1}_{r_t > \mu}. \quad (4.3)$$

The α_i parameters capture lagged shock effects, while the ϕ_i parameters capture same-day feedback effects. Asymmetry is allowed by giving negative and positive shocks different coefficients. Setting $\phi_1 = \phi_2 = 0$ gives classic GJR-GARCH; setting $\alpha_1 = \alpha_2$ and $\phi_1 = \phi_2$ gives symmetric RT-GARCH.

Unlike in standard GARCH, h_t in (4.2) is not known from yesterday's information, because it depends on the current shock. It is therefore a filtered latent volatility process. A common initialisation is $h_1 = \bar{h}$, where $\bar{h}$ is discussed in [Appendix A](#). If $\alpha_1 = \alpha_2 = 0$ and $\bar{\phi} = (\phi_1 + \phi_2)/2$, then $\bar{h} = \omega/(1 - \beta - \bar{\phi})$ when $\beta + \bar{\phi} < 1$.

4.2 Recovering h_t from r_t

Define the predictable part of the variance recursion as

$$g_{t-1} := \omega + \beta h_{t-1} + \alpha_{t-1} (r_{t-1} - \mu)^2. \quad (4.4)$$

Then we can write

$$h_t = g_{t-1} + \phi_t h_{t-1} \varepsilon_t^2, \quad (4.5)$$

and the observation equation implies

$$(r_t - \mu)^2 = h_t \varepsilon_t^2. \quad (4.6)$$

Multiplying (4.5) by h_t and using (4.6) gives the quadratic equation

$$h_t^2 - g_{t-1}h_t - \phi_t h_{t-1}(r_t - \mu)^2 = 0. \quad (4.7)$$

Choosing the positive root yields the filtering equation

$$h_t = \frac{g_{t-1} + \sqrt{g_{t-1}^2 + 4\phi_t h_{t-1}(r_t - \mu)^2}}{2}. \quad (4.8)$$

If $\phi_t = 0$, this reduces to the usual GARCH-type recursion $h_t = g_{t-1}$. Once h_t has been computed, the implied shock is

$$\varepsilon_t = \frac{r_t - \mu}{\sqrt{h_t}}. \quad (4.9)$$

To initialise the RT-GARCH recursion, set $h_0 = \bar{h}$ and $g_0 = (1 - \bar{\phi})\bar{h}$, where $\bar{\phi} = (\phi_1 + \phi_2)/2$ and $\bar{h}$ is given in Appendix A. For $t = 1$, use the filtering formula in (4.8), with g_0 and h_0 in place of g_{t-1} and h_{t-1} , to obtain h_1 ; then compute ε_1 using (4.9). For $t \geq 2$, continue recursively using (4.4) and (4.8).

5 Estimation

For each model, choose static parameters, filter h_t , compute the log-likelihood contribution $\ell_t(\boldsymbol{\theta})$, and search numerically for the parameter values that maximise the sum. Let T_{est} be the number of estimation observations. The likelihood is conditional on the chosen initial value, h_1 . Gaussian shocks are unlikely to be exactly correct for financial returns, but the Gaussian likelihood is a standard practical choice for GARCH-type models. We call this maximum likelihood (ML) estimation. The generic problem is

$$\hat{\boldsymbol{\theta}}_{\text{ML}} = \arg \max_{\boldsymbol{\theta} \in \Theta} \sum_{t=1}^{T_{\text{est}}} \ell_t(\boldsymbol{\theta}), \quad (5.1)$$

where Θ denotes the admissible parameter space and $\ell_t(\boldsymbol{\theta})$ is the log-likelihood contribution of return r_t , which is specified below for each model.

5.1 GARCH and GJR-GARCH

For GJR-GARCH, the static parameters are $\boldsymbol{\theta} = (\mu, \omega, \alpha_1, \alpha_2, \beta)'$. Symmetric GARCH imposes $\alpha_1 = \alpha_2 = \alpha$. We require $\omega > 0$, non-negative shock and persistence parameters, and the relevant stability condition: $\alpha + \beta < 1$ for GARCH and $\bar{\alpha} + \beta < 1$ for GJR-GARCH, where $\bar{\alpha} = (\alpha_1 + \alpha_2)/2$. Check that your final estimates satisfy these constraints.

For a given parameter vector, compute h_t for all t using Section 3. Conditional on $\mathcal{I}_{t-1}$, the density of r_t is

$$f(r_t | \mathcal{I}_{t-1}; \boldsymbol{\theta}) = \frac{1}{\sqrt{h_t}} p\left(\frac{r_t - \mu}{\sqrt{h_t}}\right), \quad (5.2)$$

where $p(\cdot)$ is the density of ε_t . Under the Gaussian assumption, the logarithm of this expression reads

$$\ell_t(\boldsymbol{\theta}) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(h_t) - \frac{1}{2} \frac{(r_t - \mu)^2}{h_t}.$$

In code, you usually minimise the negative log-likelihood. Since volatility likelihoods can have local optima, try several starting values and keep the best solution.

5.2 RT-GARCH and RT-GJR-GARCH

For RT-GJR-GARCH, the static parameters are $\boldsymbol{\theta} = (\mu, \omega, \alpha_1, \alpha_2, \beta, \phi_1, \phi_2)'$. Symmetric RT-GARCH imposes $\alpha_1 = \alpha_2$ and $\phi_1 = \phi_2$. After estimation, verify $\omega > 0$, $\beta, \alpha_1, \alpha_2, \phi_1, \phi_2 \geq 0$, and the unconditional variance condition in Appendix A.

The RT-GARCH density differs from the standard GARCH density because h_t depends on the current shock. With h_t filtered from (4.8), the density of r_t becomes

$$f(r_t | \mathcal{I}_{t-1}; \boldsymbol{\theta}) = \frac{\sqrt{h_t}}{h_t + \phi_t h_{t-1} (r_t - \mu)^2 / h_t} p\left(\frac{r_t - \mu}{\sqrt{h_t}}\right). \quad (5.3)$$

A derivation is given in Appendix B. Under Gaussian density $p(\cdot)$, the logarithm is

$$\ell_t(\boldsymbol{\theta}) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \frac{(r_t - \mu)^2}{h_t} + \log\left(\frac{\sqrt{h_t}}{h_t + \phi_t h_{t-1} (r_t - \mu)^2 / h_t}\right). \quad (5.4)$$

A practical strategy is to estimate GJR-GARCH first, use those estimates as starting values for $(\mu, \omega, \alpha_1, \alpha_2, \beta)$, and add small positive starting values for (ϕ_1, ϕ_2) .

6 Evaluation

We compare one-step-ahead variance forecasts with observed volatility proxies. All forecasts and targets are in squared percentage points.

6.1 Predicted volatility

Let PV_t denote the forecast of next-day variance, or $E[(r_{t+1} - \mu)^2 | \mathcal{I}_t]$. For daily returns, using r_{t+1}^2 instead of $(r_{t+1} - \mu)^2$ usually makes little difference, but the demeaned version is cleaner. For GARCH and GJR-GARCH, h_{t+1} is known at time t , so

$$PV_t = E[(r_{t+1} - \mu)^2 | \mathcal{I}_t] = h_{t+1}. \quad (6.1)$$

For RT-GARCH and RT-GJR-GARCH, h_{t+1} depends on the future shock. Define $\bar{\phi} = (\phi_1 + \phi_2)/2$, $\kappa = E[\varepsilon_t^4]$, and $g_t = \omega + \beta h_t + \alpha_t (r_t - \mu)^2$, such that $h_{t+1} = g_t + \phi_{t+1} h_t \varepsilon_{t+1}^2$. Then

$$\begin{aligned} PV_t &= E[(r_{t+1} - \mu)^2 | \mathcal{I}_t] = E[h_{t+1} \varepsilon_{t+1}^2 | \mathcal{I}_t] = E[(g_t + \phi_{t+1} h_t \varepsilon_{t+1}^2) \varepsilon_{t+1}^2 | \mathcal{I}_t] \\ &= g_t + h_t E[\phi_{t+1} \varepsilon_{t+1}^4] = g_t + h_t \bar{\phi} \kappa. \end{aligned} \quad (6.2)$$

For Gaussian shocks, $\kappa = 3$. The extra term $h_t \bar{\phi} \kappa$ appears because future volatility and the future shock are linked in RT-GARCH.

6.2 Target volatility

We use two one-day-ahead target variables:

1. Demeaned squared return:

$$TV_t^{(r)} = (r_{t+1} - \hat{\mu})^2,$$

where $\hat{\mu}$ is computed using only the relevant estimation sample. This target covers the full close-to-close interval.

2. Realised variance:

$$TV_t^{(rv)} = \text{rv5_ss}_{t+1}.$$

For the E-mini, this is also a close-to-close volatility proxy, because the futures contract trades almost continuously during the week. For stock-level extensions, check whether realised variance excludes overnight variation.

Target variable 1 is directly tied to the return series used by the GARCH-type models. Target variable 2 uses high-frequency information and is usually less noisy. Evaluate all models on the same out-of-sample dates.

6.3 Benchmark forecasts

Compare the four GARCH-type models with two benchmarks: the VIX and HAR-RV. Treating VIX_t^2 as an annualised variance, the one-day VIX forecast is

$$PV_t^{VIX} = \frac{VIX_t^2}{250}, \quad (6.3)$$

using only the VIX value available at time t .

The heterogeneous autoregressive realised variance (HAR-RV) model of [Corsi \(2009\)](#) uses daily, weekly and monthly realised-variance components:

$$RV_t^{(d)} = rv5_ss_t, \quad RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}^{(d)}, \quad RV_t^{(m)} = \frac{1}{21} \sum_{j=0}^{20} RV_{t-j}^{(d)}.$$

For one-step-ahead forecasting, estimate

$$TV_t = c + \beta^{(d)} RV_t^{(d)} + \beta^{(w)} RV_t^{(w)} + \beta^{(m)} RV_t^{(m)} + e_t, \quad (6.4)$$

where TV_t is either target variable. The HAR-RV forecast is

$$PV_t^{HAR} = \hat{c} + \hat{\beta}^{(d)} RV_t^{(d)} + \hat{\beta}^{(w)} RV_t^{(w)} + \hat{\beta}^{(m)} RV_t^{(m)}. \quad (6.5)$$

Estimate the HAR-RV model recursively or with a fixed estimation window, and report your choice. The forecast for day $t + 1$ may only use information available at time t .

6.4 Loss functions

For each model, compare PV_t with a target TV_t . Let $\mathcal{T}_{out}$ be the set of out-of-sample forecast origins and T_{out} its size. Report at least

$$MSE = \frac{1}{T_{out}} \sum_{t \in \mathcal{T}_{out}} (TV_t - PV_t)^2$$

and

$$QLIKE = \frac{1}{T_{out}} \sum_{t \in \mathcal{T}_{out}} \left\{ \log(PV_t) + \frac{TV_t}{PV_t} \right\}.$$

QLIKE is common when realised volatility is an imperfect proxy for true volatility; see [Patton \(2011\)](#). It requires $PV_t > 0$, so handle any non-positive HAR-RV forecasts explicitly and report the rule.

Compare models within the same target and loss function. The six forecast series and two targets imply at least $6 \times 2 \times 2 = 24$ loss values.

7 Roadmap

Think of the assignment as an investigation: each model gives a different explanation for why volatility and returns move together. The steps below form the baseline analysis; optional choices can be adapted if you explain them clearly. In your final report, do not refer to the case study or roadmap.

1. **Describe the data.** Plot returns, realised variance, and the VIX. Report summary statistics and discuss the main patterns.
2. **Estimate the four GARCH-type models.** Estimate GARCH, GJR-GARCH, RT-GARCH, and RT-GJR-GARCH. Report parameter estimates, log-likelihoods, and optionally information criteria.
3. **Compare in-sample fit.** Use likelihood comparisons where appropriate, especially to assess asymmetry within the classic and real-time model families. You may also compare models with and without the real-time component.

4. **Inspect filtered volatility.** Plot the filtered h_t paths and discuss whether classic and real-time models react differently in turbulent periods.
5. **Construct forecasts.** Choose and report an estimation/evaluation split, then produce one-step-ahead forecasts for GARCH, GJR-GARCH, RT-GARCH, RT-GJR-GARCH, HAR-RV, and VIX. Avoid look-ahead bias.
6. **Evaluate forecast accuracy.** Compare all forecasts against both targets using at least MSE and QLIKE (Patton, 2011). Use the same out-of-sample dates for all models. Mincer–Zarnowitz and Diebold–Mariano tests may be used for calibration and predictive-accuracy comparisons.
7. **Tell the econometric story.** Decide whether the evidence points to leverage, feedback, or both. Use the statistical analysis to support your conclusion.
8. **Optional extension.** After completing the baseline analysis, you may apply the same methods to individual stocks, exchange rates, or other futures contracts.

8 Econometric software

You may use any software package, such as EViews, MATLAB, R, Python, or Julia. A basic MATLAB demo for estimating a simple GARCH model is provided. It is only a starting point: you are responsible for adapting the code, checking your implementation, and making your results reproducible. Coding support is limited to the scheduled feedback opportunities. Please do not approach other professors for help with this assignment.

9 Proposal

The research proposal should be no more than two pages and is due on the date announced on Canvas. Use it to show your empirical plan and expected output, not to summarise the case study. Include actual output for steps one to three of the roadmap: descriptive plots and statistics, parameter estimates, log-likelihood values, and any likelihood comparisons already completed. For later steps, include planned tables or figures, even if empty. Also state your intended estimation/evaluation split. You will receive verbal feedback at the seminar announced on Canvas. Proposals that do not include output for steps one to three will receive no feedback.

10 Second meeting

A second meeting with one of the supervisors will take place on the date announced on Canvas. The supervisors will not provide written feedback, but you may submit three to five specific questions. Good questions refer to concrete choices, tables, figures, model results, or interpretation problems. Very general questions, such as whether the report is good enough, are unlikely to be useful.

11 Final report

The final report is **at most 5,000 words or 12 pages**, whichever comes first. Both limits include the cover page and references, but not appendices. Include a short abstract of at most 150 words. Put the important material in the main text; appendices should only contain supplementary material. Reports must include the group number. Late submissions will not be graded.

Your report should read as a stand-alone empirical paper. Explain the data, motivate the models, present the evidence, and take a position on whether leverage, feedback, or both matter most. Do not only report which model wins; explain what the result says about the economic mechanisms. Focus on what *you* have done. Do not refer to “the case study”; write as if your report is an independent empirical paper.

FEB22009 students may write in Dutch or English; FEB22009X, FEB22009Q and FEB22009S students must write in English. The cover page should contain the title, group number, names and student numbers of all team members, and word counts for the main text and abstract. Do not include a table of contents. The report should follow normal academic conventions, use APA-style references, and be written in $\text{\LaTeX}$.

12 Use of generative AI

You may use generative AI as a tutor, reviewer, or brainstorming partner. You may not use it to generate final answers, submit AI-written analysis as your own, produce report-ready text, final numerical results, or complete code solutions. Any use of generative AI must comply with the Erasmus Ethics code of conduct and the academic-integrity rules that apply to this course. Acknowledge any AI assistance, including the tool used, how it helped, and which parts of the work were affected. In an appendix, briefly state whether generative AI was used; if so, list the tool, the type of assistance received, and one concrete example of how you verified or revised the output.

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A Unconditional variance of RT-GARCH

This appendix derives the unconditional mean of h_t for the RT-GJR-GARCH model. This helps check stability and choose a starting value for the filter. It is mainly for students who want more detail on the stability condition. Write the model as

$$r_t - \mu = \sqrt{h_t} \varepsilon_t, \quad h_t = g_{t-1} + \phi_t h_{t-1} \varepsilon_t^2, \quad g_{t-1} = \omega + \beta h_{t-1} + \alpha_{t-1} (r_{t-1} - \mu)^2,$$

where

$$\alpha_t = \alpha_1 \mathbb{1}_{r_t \leq \mu} + \alpha_2 \mathbb{1}_{r_t > \mu}, \quad \phi_t = \phi_1 \mathbb{1}_{r_t \leq \mu} + \phi_2 \mathbb{1}_{r_t > \mu}.$$

Define

$$\bar{h} = E[h_t], \quad \bar{g} = E[g_t], \quad \bar{\alpha} = \frac{\alpha_1 + \alpha_2}{2}, \quad \bar{\phi} = \frac{\phi_1 + \phi_2}{2}.$$

Since the new shock is independent of past information and the shock distribution is symmetric,

$$\bar{h} = \bar{g} + \bar{\phi} \bar{h}, \quad \text{so} \quad \bar{g} = (1 - \bar{\phi}) \bar{h}.$$

Write $\alpha_t(r_t - \mu)^2 = \alpha_t h_t \varepsilon_t^2$ and substitute $h_t = g_{t-1} + \phi_t h_{t-1} \varepsilon_t^2$ to yield

$$\alpha_t(r_t - \mu)^2 = \alpha_t g_{t-1} \varepsilon_t^2 + \alpha_t \phi_t h_{t-1} \varepsilon_t^4.$$

Taking expectations gives

$$\mathbb{E}[\alpha_t(r_t - \mu)^2] = \bar{\alpha} \bar{g} + \frac{3}{2}(\alpha_1 \phi_1 + \alpha_2 \phi_2) \bar{h},$$

because $\mathbb{E}[\varepsilon_t^2 \mathbb{1}_{\varepsilon_t \leq 0}] = \mathbb{E}[\varepsilon_t^2 \mathbb{1}_{\varepsilon_t > 0}] = 1/2$ and $\mathbb{E}[\varepsilon_t^4 \mathbb{1}_{\varepsilon_t \leq 0}] = \mathbb{E}[\varepsilon_t^4 \mathbb{1}_{\varepsilon_t > 0}] = 3/2$ for standard normal shocks. Lagging the time index on the left and using $\bar{g} = (1 - \bar{\phi}) \bar{h}$ gives

$$\mathbb{E}[\alpha_{t-1}(r_{t-1} - \mu)^2] = \bar{\alpha}(1 - \bar{\phi}) \bar{h} + \frac{3}{2}(\alpha_1 \phi_1 + \alpha_2 \phi_2) \bar{h}.$$

Taking expectations in the volatility recursion $h_t = \omega + \beta h_{t-1} + \alpha_{t-1}(r_{t-1} - \mu)^2 + \phi_t h_{t-1} \varepsilon_t^2$ then gives

$$\bar{h} = \omega + \left\{ \beta + \bar{\phi} + \bar{\alpha}(1 - \bar{\phi}) + \frac{3}{2}(\alpha_1 \phi_1 + \alpha_2 \phi_2) \right\} \bar{h}.$$

Because $\omega > 0$, this fixed-point equation has a finite positive solution for $\bar{h}$ if and only if the term in curly brackets is less than unity,

$$\beta + \bar{\phi} + \bar{\alpha}(1 - \bar{\phi}) + \frac{3}{2}(\alpha_1 \phi_1 + \alpha_2 \phi_2) < 1.$$

Under this condition,

$$\bar{h} = \frac{\omega}{1 - \beta - \bar{\alpha} - \bar{\phi} + \bar{\alpha} \bar{\phi} - \frac{3}{2}(\alpha_1 \phi_1 + \alpha_2 \phi_2)}.$$

B Derivation of the RT-GARCH conditional density

This appendix derives equation (5.3). The key difference from standard GARCH is that, in RT-GARCH, h_t is not fixed when conditioning on $\mathcal{I}_{t-1}$. It depends on the current shock ε_t , and therefore on the observed return r_t . Conditional on $\mathcal{I}_{t-1}$, write

$$h_t(\varepsilon_t) = g_{t-1} + \phi_t h_{t-1} \varepsilon_t^2, \quad r_t(\varepsilon_t) = \mu + \sqrt{h_t(\varepsilon_t)} \varepsilon_t,$$

where g_{t-1} is defined in (4.4). After observing r_t , the corresponding value of h_t is the filtered value obtained from (4.8). In the density below, h_t should therefore be read as this filtered value $h_t(r_t)$. The inverse relation is

$$\varepsilon_t = \frac{r_t - \mu}{\sqrt{h_t}}, \quad \varepsilon_t^2 = \frac{(r_t - \mu)^2}{h_t},$$

where again $h_t = h_t(r_t)$. By the change-of-variables formula,

$$f(r_t | \mathcal{I}_{t-1}; \boldsymbol{\theta}) = p(\varepsilon_t) \left| \frac{\partial r_t}{\partial \varepsilon_t} \right|^{-1}.$$

Within each sign region, ϕ_t is constant; the point $\varepsilon_t = 0$ has probability zero under the continuous shock distribution. Since

$$\frac{\partial h_t}{\partial \varepsilon_t} = 2\phi_t h_{t-1} \varepsilon_t,$$

we get

$$\frac{\partial r_t}{\partial \varepsilon_t} = \sqrt{h_t} + \varepsilon_t \frac{1}{2\sqrt{h_t}} \frac{\partial h_t}{\partial \varepsilon_t} = \frac{h_t + \phi_t h_{t-1} \varepsilon_t^2}{\sqrt{h_t}}.$$

Substituting $\varepsilon_t^2 = (r_t - \mu)^2 / h_t$ gives

$$\left| \frac{\partial r_t}{\partial \varepsilon_t} \right| = \frac{h_t + \phi_t h_{t-1} (r_t - \mu)^2 / h_t}{\sqrt{h_t}}.$$

Therefore

$$f(r_t | \mathcal{I}_{t-1}; \boldsymbol{\theta}) = \frac{\sqrt{h_t}}{h_t + \phi_t h_{t-1} (r_t - \mu)^2 / h_t} p\left(\frac{r_t - \mu}{\sqrt{h_t}}\right),$$

where h_t depends on r_t via equation (4.8). This confirms equation (5.3). Note that if $\phi_1 = \phi_2 = 0$, such that $\phi_t = 0$, then h_t no longer depends on the current shock and this expression reduces to the standard GARCH density in (5.2).